Applied Physics Letters

RESEARCH ARTICLE | OCTOBER 11 1999

Enhanced harmonic generation in aperiodic optical superlattices

Ben-Yuan Gu; Bi-Zhen Dong; Yan Zhang; Guo-Zhen Yang

Appl. Phys. Lett. 75, 2175–2177 (1999)
https://doi.org/10.1063/1.124956

Enhanced harmonic generation in aperiodic optical superlattices

Ben-Yuan Gu[a)]
CCAST (World Laboratory), P.O. Box 8730, and Institute of Physics, Chinese Academy of Sciences, P.O. Box 603, Beijing 100080, China

Bi-Zhen Dong, Yan Zhang, and Guo-Zhen Yang
Institute of Physics, Chinese Academy of Sciences, P.O. Box 603, Beijing 100080, China

(Received 7 June 1999; accepted for publication 13 August 1999)

We find that harmonic generation can be enhanced with aperiodic optical superlattice (AOS) structures realized by inverting poled ferroelectric domains in sample. The optimal design of the AOS can be achieved with use of the simulated annealing method. The constructed AOSs can implement multiple wavelength second-harmonic generation and the coupled third-harmonic generation with an identical effective nonlinear coefficient. The simulations show that the constructed AOSs can enhance harmonic generation compared with the Fibonacci optical superlattice. The physical origin of this enhancement is ascribed to the constructive interference effect. © *1999 American Institute of Physics.* [S0003-6951(99)03841-3]

Optical parametric processes are of importance for creating laser sources at new frequencies and have been widely applied to nonlinear optics. In the traditional method, in order to get high conversion efficiency, it usually requires the so-called phase-matching (PM) condition.[1,2] This important PM condition leads to a large restriction in the choice of the naturally birefringent materials to serve in nonlinear optics. Another scheme, the so-called quasiphase matching (QPM),[3,4] has been proposed for a long time. This method substantially extends the class of materials available in nonlinear optics.[5–13] With advances of the electric poling technique, it is possible to build various optical superlattices based on ferroelectric crystals such as $LiTaO_3$, $LiNbO_3$, $KTiOPO_4$, and so on, with periodically or quasiperiodically domain-inverted structures.[5–9]

The principle of QPM is that a one-dimensional microstructural superlattice contains plentiful reciprocal vectors, thus, the combination of the reciprocal vectors with the wave vector of the beam results in a new phase-matching equation, and leads to a flexible quasiphase matching. Some researchers have devoted their works to theoretical and experimental studies on this subject.[5–11] For instance, they have reported the result of quasiphase-matched third-harmonic generation (THG) in a quasiperiodic optical superlattice,[5] and experimental realization of multiple wavelength second-harmonic generation (SHG) in a Fibonacci optical superlattice (FOS) of $LiTaO_3$ (LT).[6]

One question is naturally raised whether the FOS is the best candidate for nonlinear optical processes. A one-dimensional aperiodic optical superlattice (AOS) contains more plentiful Fourier spectral components than those in the FOS. Thus, it may be expected that the AOS can become another favorable candidate for optical parametric devices. The key question is how to construct the AOS to match the specified nonlinear optical process with efficient conversion. This is an inverse source problem in nonlinear optics.

Motivated by these works and questions, in this letter we propose to optimally construct the AOS and obtain enhanced harmonic generation. We first give the relevant description and analysis of the optimal construction problem of the AOS. We then design special AOSs that implement multiple wavelength SHG or coupled THG with identical effective nonlinear coefficient. The physical mechanism for this enhanced harmonic generation effect is that the contribution from each unit block of sample to the optical parametric processes takes constructive addition.

We consider an AOS based on LT crystals with laminar ferroelectric-domain structure. In such a material, the directions of polarization vectors in successive domains are opposite, as are the signs of nonlinear optical coefficient. However, the width of each individual domain may be different and it is determined by the specified optical parametric processes.

We first discuss the SHG process for the single wavelength case. Assume that a single laser beam with $\omega_1=\omega$ is incident from the left onto the surface of an AOS, and through the nonlinear optical process, the SHG is produced in this AOS. In order to use the largest nonlinear coefficient d_{33}, let the interfaces of each domain be parallel to the yz plane, and the propagation and the polarization directions of incident light are along the x and z axes, respectively. In the so-called small-signal approximation, the depletion loss of fundamental wave power can be negligible, and the conversion efficiency η_{SHG} from fundamental wave to second harmonic wave reads

$$\eta_{\mathrm{SHG}}=\frac{8\pi^2|d_{33}|^2 I_\omega L^2}{c\epsilon_0\lambda^2 n_{2\omega}n_\omega^2}\left|\frac{1}{L}\int_0^L dx\, e^{i(k_{2\omega}-2k_\omega)x}\tilde{d}(x)\right|^2, \quad (1)$$

where k_ω ($k_{2\omega}$) is the wave number of the fundamental (second harmonic) wave; c is the speed of light in vacuum; n_ω ($n_{2\omega}$) represents the refractive index of the fundamental (second harmonic) wave; λ is the fundamental wavelength in vacuum; and ϵ_0 is the permittivity of vacuum, L is the total length of the sample, and $\tilde{d}(x)$ only takes binary values of 1 or -1.

[a)]Electronic mail: guby@aphy.iphy.ac.cn

To solve the inverse source problem in constructing the AOS, we consider the reduced effective nonlinear coefficient for SHG defined by

$$\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)=\frac{1}{L}\left|\int_0^L dx\, e^{i[2\pi x/l_c^{(s)}(\lambda_\alpha)]}\tilde{d}(x)\right|, \tag{2}$$

where α numerates different wavelengths and $l_c^{(s)}(\lambda)=\lambda/2(n_{2\omega}-n_\omega)$ is the coherence length for SHG. Assume that the thickness of the unit block is Δx, thus the number of the blocks in sample is $N=L/\Delta x$. The position of each block is located at $x_q=q\Delta x$, for $q=0,1,2,3\ldots N-1$. We evaluate this integral Eq. (2) as

$$\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)=\frac{1}{N}\left|\mathrm{sinc}\left[\frac{\Delta x}{l_c^{(s)}(\lambda_\alpha)}\right]\times\sum_{q=0}^{N-1}\tilde{d}(x_q)e^{i[2\pi(q+0.5)\Delta x/l_c^{(s)}(\lambda_\alpha)]}\right|, \tag{3}$$

where $\mathrm{sinc}(x)=\sin(\pi x)/(\pi x)$. It is clearly seen that $\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)$ has contributions from two factors: one factor (sinc function) belongs to the unit block and strongly depends on Δx and $l_c^{(s)}(\lambda_\alpha)$; the other factor reflects the interference effect among the blocks and depends on the configuration of domains as well as the phase lagging from one block to the other block.

It is evident that the optimization construction of the AOS for the SHG can be ascribed as a search for the maximum of $\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)$ with respect to $\tilde{d}(q\Delta x)$. The maximum value condition for the second factor corresponds to the appearance of the perfectly constructive interference. Such an optimization nonlinear problem can be solved with the simulated annealing (SA) method[14,15] and then the favorable arrangement of the domain orientations of the blocks in the sample can be determined completely.

To demonstrate the effectiveness of SA for dealing with the inverse source problem concerned here, we consider a simple example for single wavelength SHG. We choose the parameters: the wavelength of incident beam is $\lambda=1.06\,\mu$m; the thickness of the basic block is $\Delta x=l_c^{(s)}/2=3.8713\,\mu$m; and the number of blocks is 2142. By using SA, we indeed obtain a perfect periodical structure with a period of $a=l_c^{(s)}$, as expected. The reduced effective nonlinear coefficient $\xi_{\mathrm{eff}}^{(s)}$ reaches its theoretical maximum value of $2/\pi=0.6366$.

We now carry out the particular construction of AOS that can achieve multiple wavelength SHG with identical $\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)=\xi^{(0)}$, where $\xi^{(0)}$ is a constant value. We choose $\Delta x=3\,\mu$m and assume that the five wavelengths of incident light are λ_α: 0.9720, 1.0820, 1.2830, 1.3640, and 1.5687 (μm). The dispersions of the refractive index of material are evaluated according to Sellmeier formula.[8] The objective function in the SA method is chosen as

$$E=\sum_\alpha\left[|\xi^{(0)}-\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)|\right]+\left[\max\{\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)\}-\min\{\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)\}\right], \tag{4}$$

where the symbol max{...}(min{...}) manifests to take their maximum (minimum) value among all the quantities including into {....}.

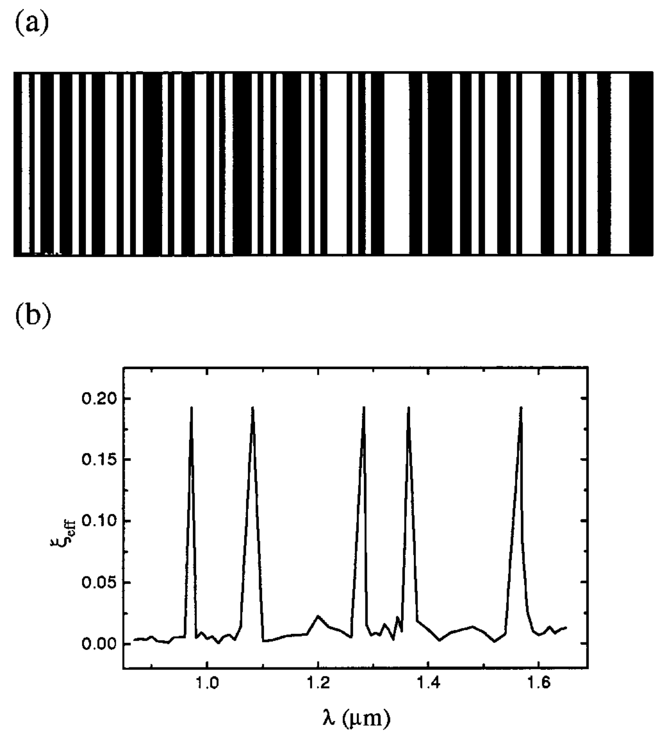

FIG. 1. Calculated results for the constructed AOS that can implement multiple wavelength SHG with large identical nonlinear coefficient: (a) a gray-scale diagram of the constructed AOS in part; (b) wavelength dependence of $\xi_{\mathrm{eff}}^{(s)}$ in the constructed AOS.

The calculated results for the sample with $L=8295\,\mu$m and $N=2765$ are displayed in Fig. 1. Figure 1(a) is a gray-scale diagram of the constructed AOS in part. The black (white) strip represents the positive (negative) domain. Figure 1(b) displays the calculated reduced effective nonlinear coefficient after scanning a wide range of wavelength. There exist five strong peaks with almost identical peak value at the predesignated wavelengths. The average $\langle\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)\rangle$ is 0.1927 and its maximal relative deviation is $\Delta\xi_{\mathrm{eff}}^{(s)}=[(\xi_{\mathrm{eff}}^{(s)})_{\max}-(\xi_{\mathrm{eff}}^{(s)})_{\min}]/\langle\xi_{\mathrm{eff}}^{(s)}\rangle=7.27\times10^{-6}$.

It is interesting to present a comparison of our constructed AOS with the FOS presented in Ref. 6. Employing the parameters in Ref. 6, we form the modulation function of $d(x)$ and calculate the integral in Eq. (2), thus, the calculated result is: $\xi_{\mathrm{eff}}^{(s)}(\lambda_\alpha)=(0.0969, 0.1860, 0.1377, 0.0825, 0.1201)$ for the wavelengths: (0.9806, 1.0893, 1.2945, 1.3785, 1.5967) (μm), respectively. The average value is 0.1246. It is clearly seen that they exhibit remarkable nonuniformity ($\Delta\xi_{\mathrm{eff}}^{(s)}=83.06\%$). Furthermore, the $\langle\xi_{\mathrm{eff}}^{(s)}\rangle$ of our constructed AOS is larger than that in the FOS presented in Ref. 6. Therefore, the SHG can be enhanced in the AOS.

The creation of the THG can take several different ways.[1,2] The more efficient way is the use of the coupled parametric processes, which are the combination of one SHG process and the other a sum-frequency generation process. In this case, the largest nonlinear optical coefficient can be available for obtaining high conversion efficiency. The coupled THG process can be analyzed by solving the coupled nonlinear equations that describe the interaction of the three fields E_ω, $E_{2\omega}$, and $E_{3\omega}$ in the AOS. Under the small signal approximation, the conversion efficiency of the THG can be expressed by

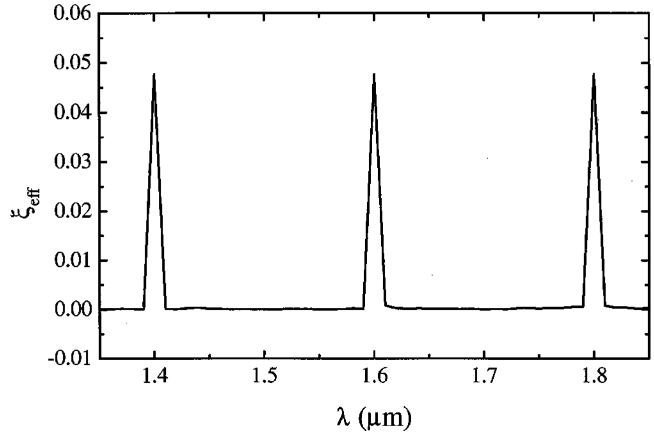

FIG. 2. Wavelength dependence of $\xi_{\text{eff}}^{(ct)}$ in the constructed AOS that implements multiple wavelength coupled THG.

$$\eta_{\text{THG}}=\frac{144\pi^4|d_{33}|^4 I_\omega^2 L^4}{c^2\epsilon_0^2\lambda^4 n_\omega^3 n_{2\omega}^2 n_{3\omega}}\xi_{\text{eff}}^{(ct)2}, \tag{5}$$

where

$$\xi_{\text{eff}}^{(ct)}(\lambda)=\frac{2}{L^2}\left|\int_0^L dx\, e^{i[2\pi x/l_c^{(t)}(\lambda_\alpha)]}\tilde{d}(x)\times\int_0^x d\xi\, e^{i[2\pi\xi/l_c^{(s)}(\lambda_\alpha)]}\tilde{d}(\xi)\right| \tag{6}$$

represents the reduced coupled effective nonlinear coefficient.

Similarly, the optimally constructing problem of the AOS structure corresponds to the maximizing $\xi_{\text{eff}}^{(ct)}(\lambda)$ with respect to function argument $\tilde{d}(x)$. We employ the SA method to design the AOS that implements the multiple wavelength coupled THG with identical effective nonlinear coefficients. The parameters are: $\lambda_\alpha=(1.40, 1.60, 1.80)\ \mu\text{m}$; $\Delta x=3\ \mu\text{m}$; $L=8067\ \mu\text{m}$; $N=2689$. The dependence of $\xi_{\text{eff}}^{(ct)}$ state on wavelength is depicted in Fig. 2. There exist three peaks with identical height. It is evident that the result satisfies the preset goal very well. The coupled effective nonlinear coefficients are almost the same for different wavelengths.

In summary, we have found the enhanced harmonic generation in AOS structures. The search for the ideal construction of the AOS corresponds to an inverse source problem in nonlinear optics which can be solved by the SA method. We carry out the model designs of the AOS structures that can achieve different optical parametric processes with higher conversion efficiency than that in the FOS. The physical mechanism for this enhancement effect of harmonic generation is ascribed to the constructive interference effect. Thus, the AOSs may provide a new kind of nonlinear optical crystals to match various practical applications in nonlinear optics.

The authors acknowledge N. B. Ming and S. N. Zhu for interesting discussions. This work was supported by the National Natural Science Foundation of China.

[1] A. Yariv and P. Yeh, *Optical Wave in Crystal* (Wiley, New York, 1984), pp. 504–551.
[2] Y. R. Shen, *The Principles of Nonlinear Optics* (Wiley, New York, 1984).
[3] N. Bloembergen and A. J. Sievers, Appl. Phys. Lett. **17**, 483 (1970).
[4] A. Armstrong, N. Bloembergen, J. Ducuing, and P. S. Pershan, Phys. Rev. **127**, 1918 (1962).
[5] S. N. Zhu, Y. Y. Zhu, and N. B. Ming, Science **278**, 843 (1997).
[6] S. N. Zhu, Y. Y. Zhu, Y. Q. Qin, H. F. Wang, C. Z. Ge, and N. B. Ming, Phys. Rev. Lett. **78**, 2752 (1997).
[7] S. N. Zhu, Y. Y. Zhu, Z. Y. Zhang, H. Shu, H. F. Wang, J. F. Hong, C. Z. Ge, and N. B. Ming, J. Appl. Phys. **77**, 5481 (1995).
[8] J.-P. Meyn and M. M. Fejer, Opt. Lett. **22**, 1214 (1997).
[9] G. D. Miller, R. G. Batchko, W. M. Tulloch, D. R. Weise, M. M. Fejer, and R. L. Ber, Opt. Lett. **22**, 1834 (1997).
[10] D. Feng, N. B. Ming, J. H. Hong, Y. S. Yang, J. S. Zhu, Z. Yang, and Y. N. Wang, Appl. Phys. Lett. **37**, 607 (1980).
[11] Y. H. Xue, N. B. Ming, J. S. Zhu, and D. Feng, Acta Phys. Sin. **32**, 1515 (1983) [Chin. Phys. **4**, 554 (1984)].
[12] D. E. Thompson, J. D. McMullen, and D. B. Anderson, Appl. Phys. Lett. **29**, 113 (1973).
[13] C. F. Deway and L. O. Hocker, Appl. Phys. Lett. **26**, 442 (1975).
[14] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, Science **220**, 671 (1983).
[15] S. Kirkpatrick, J. Stat. Phys. **34**, 975 (1983).